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Integration

1.1 Integration Respect to a Measure

Theorem 1.1.1 ► Monotone Convergence Theorem

Suppose $(X, \mathcal{S}, \mu)$ is a measure space and $0 \leq f_1 \leq f_2 \leq \dots$ is an increasing sequence of $\mathcal{S}$ -measurable functions. Define $f : X \rightarrow [0, \infty)$ by

$$f(x) = \lim_{k \rightarrow \infty} f_k(x).$$

Then

$$\lim_{k \rightarrow \infty} \int f_k d\mu = \int f d\mu.$$

Proof. f is $\mathcal{S}$ -measurable as the pointwise limitation of $\{f_k\}$. Since $f_k \leq f$ for all k , we have $\int f_k d\mu \leq \int f d\mu$. Thus

$$\lim_{k \rightarrow \infty} \int f_k d\mu \leq \int f d\mu$$

To show the converse, take arbitrary simple function $s = \sum_{j=1}^n c_j \chi_{A_j}$ with standard representation such that $f \geq s$. We fix a $t \in (0, 1)$, and define

$$E_k := \left\{ x \in X : f_k(x) \geq t \sum_{j=1}^n c_j \chi_{A_j}(x) \right\}$$

Then E_k s are measurable and

$$E_1 \subseteq E_2 \subseteq \dots, \quad \bigcup_{k=1}^{\infty} E_k = X$$

Then it follows that $\mu(A_j) = \lim_{k \rightarrow \infty} \mu(A_j \cap E_k)$ for each j . We have

$$\int f_k d\mu \geq \int t \sum_{j=1}^n c_j \chi_{A_j \cap E_k} d\mu \geq t \sum_{j=1}^n c_j \mu(A_j \cap E_k)$$

By taking $k \rightarrow \infty$, we get

$$\lim_{k \rightarrow \infty} \int f_k d\mu \geq t \sum_{j=1}^n c_j \mu(A_j)$$

Taking the supremum of the inequality above over all simple functions s , we get

$$\lim_{k \rightarrow \infty} \int f_k d\mu \geq t \int f d\mu$$

Finally, taking the limit of t increasing to 1 shows that

$$\lim_{k \rightarrow \infty} \int f_k d\mu \geq \int f d\mu$$

□

Lemma 1.1.2 ► Fatou's lemma

If $\{f_k\}_{k=1}^{\infty}$ is a sequence of nonnegative μ -measurable functions, then

$$\int_X \liminf_{k \rightarrow \infty} f_k d\mu \leq \liminf_{k \rightarrow \infty} \int_X f_k d\mu.$$

Proof. Let $g_k(x) = \inf_{n \geq k} f_n(x)$. Then $0 \leq g_1 \leq g_2 \leq \dots$ is an increasing sequence of $\mathcal{S}$ -measurable functions. Then

$$g(x) = \liminf_{k \rightarrow \infty} f_k$$

It follows from Monotone Convergence Theorem that

$$\int_X \liminf_{k \rightarrow \infty} f_k d\mu = \lim_{k \rightarrow \infty} \int_X g_k d\mu = \lim_{k \rightarrow \infty} \int_X g_k d\mu$$

Since for each g_k , we have

$$g_k(x) = \inf_{n \geq k} f_n(x) \leq f_k(x)$$

then

$$\int_X g_k d\mu \leq \int_X f_k d\mu, \quad \forall k \geq 1$$

By taking the $\liminf$ relative to k both sides, we have

$$\lim_{k \rightarrow \infty} \int_X g_k d\mu = \liminf_{k \rightarrow \infty} \int_X g_k d\mu \leq \liminf_{k \rightarrow \infty} \int_X f_k d\mu.$$

Thus

$$\int_X \liminf_{k \rightarrow \infty} f_k d\mu \leq \liminf_{k \rightarrow \infty} \int_X f_k d\mu$$

□

1.2 Limits of Integrations

1.2.1 Bounded Convergence Theorem

Proposition 1.2.1 ► bounding an integral

Suppose $(X, \mathcal{S}, \mu)$ is a measure space, $E \in \mathcal{S}$, and $f : X \rightarrow [-\infty, \infty]$ is a function such that $\int_E f d\mu$ is defined. Then

$$\left| \int_E f d\mu \right| \leq \mu(E) \sup_E |f|.$$

Proof. Let $c = \sup_E |f|$, then

$$\begin{aligned}
 \left| \int_E f \, d\mu \right| &= \left| \int \chi_E f \, d\mu \right| \\
 &\leq \int |\chi_E f| \, d\mu \\
 &\leq \int \chi_E c \, d\mu \\
 &= \mu(E) c = \mu(E) \sup_E |f|
 \end{aligned}$$

□

1.2.2 Dominated Convergence Theorem

Proposition 1.2.2 ► integrals on small sets are small

Suppose $(X, \mathcal{S}, \mu)$ is a measure space, $g : X \rightarrow [0, \infty]$ is $\mathcal{S}$ -measurable, and $\int g \, d\mu < \infty$. Then for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\int_B g \, d\mu < \varepsilon$$

for every set $B \in \mathcal{S}$ such that $\mu(B) < \delta$.

Remark.

- $g \in L^1(\mu)$ induces a finite measure

$$\nu(B) = \int_B g \, d\mu, \quad B \in \mathcal{S}$$

- For each $\varepsilon > 0$, there exists $\delta > 0$ such that $\nu(B) < \varepsilon$ whenever $\mu(B) < \delta$.

Proof. Since $\int g \, d\mu < \infty$, for each $\varepsilon > 0$, there exists a simple function $h : X \rightarrow [0, \infty]$ such that $h \leq g$,

$$0 \leq \int g \, d\mu - \int h \, d\mu < \frac{\varepsilon}{2}$$

Suppose that $c = \sup_X h$, which is finite. Then for $\delta = \frac{\varepsilon}{2c}$, we have for each $B \in \mathcal{S}$ such that $\mu(B) < \delta$, there is

$$\int_B h \, d\mu < \mu(B) c < \frac{\varepsilon}{2}$$

$$\int_B g \, d\mu \leq \int_B h \, d\mu + \frac{\varepsilon}{2} < \varepsilon$$

□

Proposition 1.2.3 ▶ integrable functions live mostly on sets of finite measure

Suppose $(X, \mathcal{S}, \mu)$ is a measure space, $g : X \rightarrow [0, \infty]$ is $\mathcal{S}$ -measurable, and $\int g \, d\mu < \infty$. Then for every $\varepsilon > 0$, there exists $E \in \mathcal{S}$ such that $\mu(E) < \infty$ and

$$\int_{X \setminus E} g \, d\mu < \varepsilon.$$

Proof. Let $h : X \rightarrow [0, \infty]$ be a simple function such that $0 \leq h \leq g$,

$$\int g \, d\mu - \int h \, d\mu < \frac{\varepsilon}{2}$$

Define

$$E = h^{-1}((0, \infty))$$

It follows from the definition of the simple function and its integrability that $\mu(E) < \infty$. Furthermore,

$$\int_{X \setminus E} h \, d\mu = \int_{X \setminus E} 0 \cdot d\mu = 0$$

Thus

$$\int_{X \setminus E} g \, d\mu \leq \frac{\varepsilon}{2} + \int_{X \setminus E} h \, d\mu < \varepsilon$$

□

Theorem 1.2.4 ▶ Dominated Convergence Theorem

Suppose $(X, \mathcal{S}, \mu)$ is a measure space, $f : X \rightarrow [-\infty, \infty]$ is $\mathcal{S}$ -measurable, and $f_1, f_2, \dots$ are $\mathcal{S}$ -measurable functions from X to $[-\infty, \infty]$ such that

$$\lim_{k \rightarrow \infty} f_k(x) = f(x)$$

for almost every $x \in X$. If there exists an $\mathcal{S}$ -measurable function $g : X \rightarrow [0, \infty]$ such that

$$\int g d\mu < \infty \quad \text{and} \quad |f_k(x)| \leq g(x)$$

for every $k \in \mathbb{Z}^+$ and almost every $x \in X$, then

$$\lim_{k \rightarrow \infty} \int f_k d\mu = \int f d\mu.$$

Proof sketch.

将积分差先控制在 g 几乎落在其上的有限测度集 E , 再将 E 上的积分分别控制在一个小测度集 B_ε 和使得 f_k 一致收敛的大测度集 $E \setminus B_\varepsilon$ 上.

Proof. For each $E \in \mathcal{S}$, and $k \in \mathbb{Z}^+$,

$$\begin{aligned} \left| \int f_k d\mu - \int f d\mu \right| &\leq \int_E |f_k - f| d\mu + \left| \int_{X \setminus E} f_k d\mu \right| + \left| \int_{X \setminus E} f d\mu \right| \\ &\leq \int_E |f_k - f| d\mu + 2 \int_{X \setminus E} g d\mu \end{aligned}$$

From proposition 1.2.3, for every $\varepsilon > 0$, there exists $E_\varepsilon \in \mathcal{S}$ such that $\mu(E_\varepsilon) < \infty$ and

$$\int_{X \setminus E_\varepsilon} g d\mu < \frac{\varepsilon}{4}$$

From proposition 1.2.2, there exists $\delta > 0$ such that for each $F \in \mathcal{S}$, $\mu(F) < \delta$, we have

$$\int_F g d\mu < \frac{\varepsilon}{8}$$

The Egorov's theorem implies that there exists measurable set $B_\varepsilon \in \mathcal{S}$ with

$\mu(E_\varepsilon \setminus B_\varepsilon) < \delta$ such that f_k converges uniformly to f on B_ε . Then we have

$$\lim_{k \rightarrow \infty} \int_{B_\varepsilon} |f_k - f| \, d\mu \leq \lim_{k \rightarrow \infty} \mu(B_\varepsilon) \sup_{B_\varepsilon} |f_k - f| = 0$$

Then there exists N such that

$$\int_{B_\varepsilon} |f_k - f| \, d\mu < \frac{\varepsilon}{4}, \quad k \geq N$$

Furthermore, for each k we have

$$\int_{E_\varepsilon \setminus B_\varepsilon} |f_k - f| \, d\mu \leq 2 \int_{E_\varepsilon \setminus B_\varepsilon} g \, d\mu < \frac{\varepsilon}{4}$$

Therefore

$$\int_{E_\varepsilon} |f_k - f| \, d\mu \leq \int_{E_\varepsilon \setminus B_\varepsilon} |f_k - f| \, d\mu + \int_{B_\varepsilon} |f_k - f| \, d\mu < \frac{\varepsilon}{2}, \quad k \geq N$$

then it follows that

$$\left| \int f_k \, d\mu - \int f \, d\mu \right| \leq \int_{E_\varepsilon} |f_k - f| \, d\mu + 2 \int_{X \setminus E_\varepsilon} g \, d\mu < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon, \quad k \geq N$$

That is

$$\lim_{k \rightarrow \infty} \int f_k \, d\mu = \int f \, d\mu$$

□

1.3 Integrable Functions and Continuous Functions

Definition 1.3.1

Denote $C_c(\mathbb{R}^n)$ by the collection of compactly supported continuous functions.

Theorem 1.3.2

Let E be a measurable set. $C_c(\mathbb{R}^n)$ is dense in $L(E)$. That is if $f \in L(E)$, then for each $\varepsilon > 0$, there exists $g \in C_c(\mathbb{R}^n)$ such that

$$\int_E |f - g| \, d\mu < \varepsilon$$

Remark.

- We can choose g such that $\sup_{x \in \mathbb{R}^n} |g| \leq \sup_{x \in E} |f|$

Proof. Let $f \in L(E)$, then for each $\varepsilon > 0$, there exists a simple function φ with compact support on E such that

$$\int_E |f - \varphi| \, d\mu < \frac{\varepsilon}{2}$$

Let $F = \text{supp } \varphi$, then F is bounded. Set $|\varphi(x)| \leq M, \forall x \in F$. From corollary ??, there is a $g \in C_c(\mathbb{R}^n)$ with such that

$$m(\{x \in F : |g - \varphi| > 0\}) < \frac{\varepsilon}{4M}, \quad \sup_{x \in \mathbb{R}^n} |g| \leq M$$

Then

$$\int_E |g - \varphi| \, d\mu = \int_{\{|g - \varphi| > 0\}} |g - \varphi| \, d\mu \leq 2Mm(\{|g - \varphi| > 0\}) < \frac{\varepsilon}{2}$$

We have

$$\int_E |f - g| \, d\mu \leq \int_E |f - \varphi| \, d\mu + \int_E |g - \varphi| \, d\mu < \varepsilon$$

□

Corollary 1.3.3

Let $f \in L(E)$, then there exists a sequence $\{g_k(x)\}$ consists of compactly supported continuous functions on $\mathbb{R}^n$, such that

•

$$\lim_{k \rightarrow \infty} \int_E |f - g_k| \, d\mu = 0$$

•

$$\lim_{k \rightarrow \infty} g_k(x) = f(x), \quad a.e. x \in E$$

Remark.

- g_k can be chosen to satisfy $\sup_{x \in E} |g_k| \leq \sup_{x \in E} |f|$

Proposition 1.3.4

Let $f \in L^1(\mathbb{R}^n)$. If for any $\varphi \in C_c(\mathbb{R}^n)$, there is

$$\int_{\mathbb{R}^n} f \cdot \varphi \, d\mu = 0$$

Then $f = 0$, a.e. $x \in \mathbb{R}^n$.

Proof. Let $E = \{f > 0\}$. There is a sequence of compactly supported continuous functions $\{\varphi_k\}$ such that

$$\int_{\mathbb{R}^n} |\chi_E - \varphi_k| \, d\mu = 0$$

$$\lim_{k \rightarrow \infty} \varphi_k(x) = \chi_E(x), \quad a.e. x \in \mathbb{R}^n,$$

$$|\varphi_k| \leq 1, \forall k$$

Since $|\varphi_k f| \leq |f|$, we have from dominated convergence theorem that

$$\begin{aligned} 0 < \int_E f \, d\mu &= \int_{\mathbb{R}^n} \chi_E \cdot f \, d\mu \\ &= \lim_{k \rightarrow \infty} \int_{\mathbb{R}^n} f \cdot \varphi_k \, d\mu \end{aligned} \tag{1.1}$$

But $\lim_{k \rightarrow \infty} \int_{\mathbb{R}^n} f \cdot \varphi_k \, d\mu = 0$ for each k , which is a contradiction. □

Proposition 1.3.5

Let $f \in L^1([a, b])$, if for each differentiable function φ with $\text{supp } \varphi \subseteq$

(a, b) , there is

$$\int_{[a,b]} f(x) \varphi'(x) d\mu = 0$$

Then $f(x) = c$, a.e. $x \in [a, b]$.

Theorem 1.3.6

If $f \in L^1(\mathbb{R}^n)$, then

$$\lim_{h \rightarrow 0} \int_{\mathbb{R}^n} |f(x+h) - f(x)| dx = 0$$

Proof sketch.

紧支连续函数是一致连续的, 因此 f 和紧支的一致连续函数仅仅相差一个积分很小的函数. 而紧支的一致连续函数的积分自动是关于平移连续的.

Proof. For each $\varepsilon > 0$, we can decompose f into $f = f_1 + f_2$, where $f_1 \in C_c(\mathbb{R}^n)$, and

$$\int_{\mathbb{R}^n} |f_2| d\mu < \frac{\varepsilon}{4}$$

f_1 is uniformly continuous, there exists $\delta > 0$ such that for each h with $|h| < \delta$

$$\int_{\mathbb{R}^n} |f_1(x+h) - f_1(x)| d\mu = \int_{\text{supp } f_1} |f_1(x+h) - f_1(x)| d\mu < \frac{\varepsilon}{2}$$

Then we have

$$\begin{aligned} \int_{\mathbb{R}^n} |f(x+h) - f(x)| d\mu &\leq \int_{\mathbb{R}^n} |f_1(x+h) - f_1(x)| d\mu + \int_{\mathbb{R}^n} |f_2(x+h) - f_2(x)| d\mu \\ &\leq \int_{\mathbb{R}^n} |f_1(x+h) - f_1(x)| d\mu + 2 \int_{\mathbb{R}^n} |f_2(x)| d\mu \\ &< \varepsilon \end{aligned} \tag{1.2}$$

Thus

$$\lim_{h \rightarrow 0} \int_{\mathbb{R}^n} |f(x+h) - f(x)| dx = 0$$

□

Proposition 1.3.7

Let $E \subseteq \mathbb{R}^n$ be a measurable set. If $f \in L(E)$, then there exists a sequence of measurable set functions with compact support, say $\{\varphi_k\}$ such that

•

$$\lim_{k \rightarrow \infty} \varphi_k(x) = f(x), \quad a.e. x \in E$$

•

$$\lim_{k \rightarrow \infty} \int_E |f - \varphi_k| d\mu = 0$$

Proof. For each $\varepsilon > 0$, there exists $g \in C_c(\mathbb{R}^n)$ such that

$$\int_E |f - g| d\mu < \frac{\varepsilon}{2}$$

We suppose that $\text{supp } g \subseteq I$, where I is a closed cube. Since g is uniformly continuous, we can separate I into 2^m small cubes I_k such that

$$|g(x) - g(\xi_k)| < \frac{\varepsilon}{2\mu(I)}, \quad k = 1, 2, \dots, 2^m$$

where ξ_k is a point in I_k . Define $\varphi(x) = \sum_{k=1}^{2^m} g(\xi_k) \chi_{I_k}(x)$, then note that $\sum_{k=1}^{2^m} \mu(I_k) = \mu(I)$, we have

$$\int_I |g - \varphi| dx < \sum_{k=1}^{2^m} \mu(I_k) \frac{\varepsilon}{2\mu(I)} = \frac{\varepsilon}{2}$$

Take $\varphi_k = \frac{1}{k}$, $k = 1, 2, \dots$, we get a sequence $\{\varphi_k\}$ such that

$$\lim_{k \rightarrow \infty} \int_E |f - \varphi_k| dx = 0$$

Then $\{\varphi_k\}$ convergent to f in measure, the proposition follows from Riesz Theorem. □

1.4 Riemann and Lebesgue Integration

Definition 1.4.1

By a **partition** $\mathcal{P}$ of $[a, b]$, we mean a finite set of points $\{x_i\}_{i=0}^m$ such that $a = x_0 < x_1 < \cdots < x_m = b$. Let

$$\|\mathcal{P}\| := \max\{x_i - x_{i-1} : 1 \leq i \leq m\}$$

For each $i \in \{1, \dots, m\}$ let x_i^* be an arbitrary point of the interval $[x_{i-1}, x_i]$. A bounded function $f : [a, b] \rightarrow \mathbb{R}$ is **Riemann integrable** if

$$\lim_{\|\mathcal{P}\| \rightarrow 0} \sum_{i=1}^m f(x_i^*) (x_i - x_{i-1})$$

exists, in which case the value is the Riemann integral of f over $[a, b]$, which we will denote by

$$(R) \int_a^b f(x) \, dx$$

Lemma 1.4.2

Given a partition $\mathcal{P} = \{x_i\}_{i=0}^m$ of $[a, b]$, set

$$U(\mathcal{P}) = \sum_{i=1}^m \left[\sup_{x \in [x_{i-1}, x_i]} f(x) \right] (x_i - x_{i-1})$$

$$L(\mathcal{P}) = \sum_{i=1}^m \left[\inf_{x \in [x_{i-1}, x_i]} f(x) \right] (x_i - x_{i-1})$$

Then

$$L(\mathcal{P}) \leq \sum_{i=1}^m f(x_i^*) (x_i - x_{i-1}) \leq U(\mathcal{P})$$

for every choice of the x_i^* .

We see that a bounded function f is Riemann integrable if and only if

$$\lim_{\|\mathcal{P}\| \rightarrow 0} (U(\mathcal{P}) - L(\mathcal{P})) = 0$$

in which case

$$(R) \int_a^b f \, dx = \lim_{\|\mathcal{P}\| \rightarrow 0} U(\mathcal{P}) = \lim_{\|\mathcal{P}\| \rightarrow 0} L(\mathcal{P})$$

Lemma 1.4.3

Suppose that $\mathcal{P} = \{x_i\}_{i=1}^m$ is a partition of $[a, b]$, $z \in [a, b] \setminus \mathcal{P}$, and $\mathcal{Q} = \mathcal{P} \cup \{z\}$. Thus $\mathcal{P} \subseteq \mathcal{Q}$, $\mathcal{Q}$ is called a refinement of $\mathcal{P}$. We have

$$L(\mathcal{P}) \leq L(\mathcal{Q}) \leq U(\mathcal{Q}) \leq U(\mathcal{P})$$

Theorem 1.4.4

If $f : [a, b] \rightarrow \mathbb{R}$ is a bounded Riemann integrable function, then f is Lebesgue integrable and

$$(R) \int_a^b f(x) \, dx = \int_{[a,b]} f \, d\lambda$$

Proof. Let $\{\mathcal{P}_k\}_{k=1}^\infty$ be a sequence of partitions of $[a, b]$ such that $\mathcal{P}_k \subseteq \mathcal{P}_{k+1}$ and $\|\mathcal{P}_k\| \rightarrow 0$ as $k \rightarrow \infty$. Write $\mathcal{P}_k = \{x_j^k\}_{j=0}^{m_k}$. For each k , define l_k, u_k by

$$l_k(x) = \sum_{j=1}^{m_k} \chi_{[x_{j-1}, x_j)} \inf_{t \in [x_{j-1}, x_j]} f(t)$$

$$u_k(x) = \sum_{j=1}^{m_k} \chi_{[x_{j-1}, x_j)} \sup_{t \in [x_{j-1}, x_j]} f(t)$$

Then

$$\int_{[a,b]} l_k \, d\lambda = L(\mathcal{P}_k) \leq U(\mathcal{P}_k) = \int_{[a,b]} u_k \, d\lambda$$

Since l_k is increasing, u_k is decreasing, the limit

$$l(x) := \lim_{k \rightarrow \infty} l_k(x), \quad u(x) := \lim_{k \rightarrow \infty} u_k(x)$$

exists and are measurable and $l \leq f \leq u$. Since f is Riemann measurable, it follows from Dominated Convergence Theorem ($(u_1 - l_1)$ as a dominating function) that

$$\int_{[a,b]} (u - l) d\lambda = \lim_{k \rightarrow \infty} \int_{[a,b]} (u_k - l_k) d\lambda = \lim_{k \rightarrow \infty} (U(\mathcal{P}_k) - L(\mathcal{P}_k)) = 0$$

Thus $u = f = l$, a.e. $x \in [a, b]$. Then

$$\int_{[a,b]} f d\lambda = \lim_{k \rightarrow \infty} \int_{[a,b]} u_k d\lambda = \lim_{k \rightarrow \infty} U(\mathcal{P}_k) = (R) \int_a^b f(x) dx$$

□

Theorem 1.4.5

A bounded function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable if and only if f is continuous λ -a.e. on $[a, b]$.

Proof. Suppose f is Riemann integrable and let $\{\mathcal{P}_k\}$ be a sequence of partitions of $[a, b]$ such that $\mathcal{P}_k \subseteq \mathcal{P}_{k+1}$ and $\lim_{k \rightarrow \infty} \|\mathcal{P}_k\| = 0$. Let l_k, u_k as in the proof of the above theorem. If $x \in [a, b] \setminus N$, $l(x) = u(x)$ and $\varepsilon > 0$, then there is an integer k such that

$$u_k(x) - l_k(x) < \varepsilon$$

Let $\mathcal{P}_k = \{x_j^k\}_{j=0}^{m_k}$. Then $x \in [x_{j-1}^k, x_j^k)$ for some $j \in \{1, \dots, m_k\}$. For every $y \in (x_{j-1}^k, x_j^k)$

$$|f(y) - f(x)| \leq u_k(x) - l_k(x) < \varepsilon$$

Thus f is continuous at x . Since $l(x) = u(x)$ for λ -a.e. $x \in [a, b]$ and $\lambda(N) = 0$, we see that f is continuous at λ -a.e. point of $[a, b]$.

Now suppose f is bounded, $N \subseteq [a, b]$ with $\lambda(N) = 0$, and f is continuous at each point of $[a, b] - N$. Let $\{\mathcal{P}_k\}$ be a sequence of partitions of $[a, b]$ such that $\lim_{k \rightarrow \infty} \|\mathcal{P}_k\| = 0$. For each k , define l_k, u_k as above. If $x \in [a, b] - N$

and $\varepsilon > 0$, then there exists a $\delta > 0$ such that

$$|f(x) - f(y)| < \frac{\varepsilon}{2}$$

whenever $|y - x| < \delta$. There is a k_0 such that $\|\mathcal{P}_k\| < \frac{\delta}{2}$ whenever $k > k_0$, thus

$$u_k(x) - l_k(x) \leq 2 \sup \{|f(x) - f(y)| : |y - x| < \delta\} < \varepsilon$$

whenever $k > k_0$. Thus

$$\lim_{k \rightarrow \infty} (u_k(x) - l_k(x)) = 0$$

for each $x \in [a, b] \setminus N$. By the dominated convergence theorem

$$\lim_{k \rightarrow \infty} (U(\mathcal{P}_k) - L(\mathcal{P}_k)) = \lim_{k \rightarrow \infty} \int_{[a, b]} (u_k - l_k) d\lambda = 0$$

□

1.5 Improper Integrals

Definition 1.5.1

Let $a \in \mathbb{R}$ and let $f : [a, \infty) \rightarrow \mathbb{R}$ be a function that is Riemann integrable on each subinterval of $[a, \infty)$. The improper integral of f is defined as

$$(I) \int_a^\infty f(x) dx := \lim_{b \rightarrow \infty} (R) \int_a^b f(x) dx. \quad (1.3)$$

If the limit is finite, we say that the improper integral of f exist. We have the following result.

Theorem 1.5.2

et $f : [a, \infty) \rightarrow \mathbb{R}$ be a nonnegative function that is Riemann integrable on each subinterval of $[a, \infty]$. Then

$$\int_{[a, \infty)} f d\lambda = \lim_{b \rightarrow \infty} (R) \int_a^b f dx. \quad (1.4)$$

Thus, f is Lebesgue integrable on $[a, \infty)$ if and only if the improper integral $(I) \int_a^\infty f(x) dx$ exists. Moreover, in this case, $\int_{[a, \infty)} f(x) d\lambda = (I) \int_a^\infty f(x) dx$.

Proof. Let $b_n, n = 1, 2, 3, \dots$ be any sequence with $b_n \rightarrow \infty, b_n > a$. We define $f_n = f \chi_{[a, b_n]}$. The monotone Convergence theorem yields

$$\int_{[a, \infty)} f d\lambda = \lim_{n \rightarrow \infty} \int_{[a, \infty)} f_n d\lambda = \lim_{n \rightarrow \infty} \int_{[a, b_n]} f d\lambda$$

Since f is Riemann integrable on each interval $[a, b_n]$, we have

$$\int_{[a, b_n]} f d\lambda = (R) \int_a^{b_n} f dx$$

and we conclude that

$$\int_{[a, \infty)} f d\lambda = \lim_{n \rightarrow \infty} (R) \int_a^{b_n} f dx = \lim_{b \rightarrow \infty} (R) \int_a^b f dx$$

The last equality follows from the nonnegativity of f . □

Theorem 1.5.3

Let $f : [a, \infty) \rightarrow \mathbb{R}$ be Riemann integrable on every subinterval of $[a, \infty)$. Then f is Lebesgue integrable if and only if the improper integral $(I) \int_a^\infty |f(x)| dx$ exists. Moreover, in this case,

$$\int_{[a, \infty)} f d\lambda = (I) \int_a^\infty f(x) dx. \quad (1.5)$$

Proof. Note that $f^+ = \frac{|f|+f}{2}, f^- = \frac{|f|-f}{2}$ is Riemann integrable, if f is Riemann integrable. And we have

$$(R) \int_a^{b_n} |f| dx = (R) \int_a^{b_n} f^+ dx + (R) \int_a^{b_n} f^- dx$$

The remaining is straightforward to check.

